PROJECT AND TEAM INFORMATION

Project Title

Intelligent Traffic Routing and Dynamic Congestion Simulation

Student / Team Information

Team Name: DSCPP – III – 2026 – T355

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PROPOSAL DESCRIPTION

1. Motivation

Urban traffic congestion is a major problem that leads to increased travel time, fuel consumption, and inefficient utilization of road networks. Conventional route selection may rely on static or periodically updated information and may not respond effectively to sudden changes such as heavy congestion, accidents, or road closures.

This project aims to develop an **Intelligent Traffic Routing and Dynamic Congestion Simulation** system that determines efficient routes while adapting to changing road conditions. The road network will be represented as a **weighted graph**, where intersections are modelled as nodes and roads as edges. Dynamic traffic conditions will modify the weights of the edges, enabling the system to identify alternative routes and perform route recalculation when conditions change. The **A*** pathfinding algorithm, supported by appropriate data structures such as graphs and priority queues, will be used for efficient route selection.

The project will provide an interactive interface for visualizing the road network, traffic conditions, selected routes, and simulated events such as congestion, accidents, and road closures. It will demonstrate the practical application of **Data Structures, Algorithms, and Object-Oriented Programming in C++** to a real-world transportation problem.

In a further phase, **AI/ML-based features may be incorporated** to predict traffic conditions using historical or simulated traffic data. These predictions could be used to dynamically estimate future road congestion and improve route selection and rerouting decisions.

2. State of the Art / Current Solution

Modern navigation and transportation systems use digital maps, geographical data, real-time or periodically updated traffic information, and route optimization algorithms to provide users with efficient routes. Applications such as navigation platforms can consider changing traffic conditions and suggest alternative routes when congestion or road disruptions occur.

However, such real-world systems depend on extensive infrastructure, large-scale geographical databases, GPS and location services, real-time traffic data, and external APIs. Replicating this infrastructure is beyond the scope of this academic project.

Instead, our project will develop a **controlled and interactive traffic simulation environment** in which a road network, traffic congestion, accidents, and road closures can be modelled and modified dynamically. The system will focus on demonstrating the underlying **Data Structures and Algorithms** used for route optimization. A weighted graph with an adjacency-list representation will model the road network, while a priority queue will support the **A*** pathfinding algorithm for efficient route selection. Dynamic changes in road conditions will update edge weights and trigger route recalculation.

In a future phase, AI/ML techniques may also be incorporated to predict traffic conditions and further improve dynamic route planning.

3. Project Goals and Milestones

Goals

The primary goals of the project are:

- Develop an interactive simulation of a dynamic road network.
- Model intersections and roads using a **weighted graph with an adjacency-list representation**.
- Implement the **A*** pathfinding algorithm in C++ for efficient route selection.
- Implement and utilize a **priority queue/min-heap** to support the routing process.
- Develop a dynamic traffic model that modifies road weights based on changing traffic conditions.
- Implement automatic **route recalculation and rerouting** in response to congestion, accidents, and road closures.
- Design an interactive UI for visualizing the road network, traffic conditions, vehicles, and calculated routes.
- Provide route information such as distance, estimated travel cost/time, and nodes explored.
- Evaluate the efficiency and performance of the implemented Data Structures and Algorithms.
- Keep the system extensible for potential **AI/ML-based traffic prediction and optimization** in a future phase.

Milestones

- **Milestone 1:** Requirement analysis, problem definition, and overall system design.
- **Milestone 2:** Design and implement the road-network graph, adjacency list, and supporting data structures in C++.
- **Milestone 3:** Implement and validate the A* routing algorithm with route reconstruction.
- **Milestone 4:** Develop the dynamic traffic model and implement congestion, accident, and road-closure events.
- **Milestone 5:** Implement dynamic edge-weight updates and automatic route recalculation.
- **Milestone 6:** Develop the interactive UI and road-network/map visualization.
- **Milestone 7:** Integrate the C++ routing engine, traffic simulator, and UI, followed by system testing and performance analysis.
- **Milestone 8:** Optimize the system, finalize documentation, and prepare the final demonstration and presentation.

4. Project Approach

The system will be developed as an interactive traffic simulation consisting of a **user interface, traffic simulation module, and C++-based routing engine**. The interface will allow users to interact with the road network, select source and destination locations, modify traffic conditions, and visualize the calculated routes.

The road network will be modelled as a **weighted graph using an adjacency-list representation**, where intersections are represented as vertices and roads as edges. Each road will maintain relevant attributes such as distance, traffic level, travel cost, and availability. These attributes will be used to determine the dynamic cost of travelling through each road.

The routing engine will implement the **A*** pathfinding algorithm. A priority queue/min-heap will be used to efficiently select the most promising node during the search. The algorithm will evaluate each node using the function:

f(n) = g(n) + h(n)

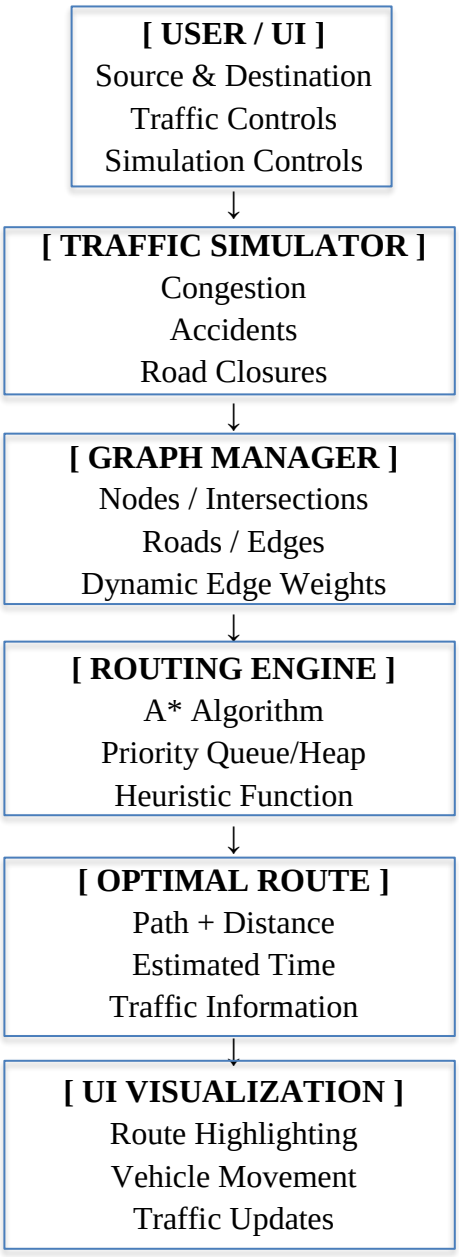
where **g(n)** represents the cost of reaching the current node and **h(n)** represents the estimated cost from the current node to the destination.

The traffic simulation module will dynamically modify road weights based on changing traffic conditions. Events such as increased congestion, accidents, and road closures will be simulated through the interface. When a change affects the selected route, the system will update the graph and execute A* again to determine an alternative optimal route.

The UI will provide **interactive road-network visualization, source/destination selection, traffic controls, route highlighting, vehicle simulation, and route statistics** such as distance, estimated travel time, and nodes explored.

The project will primarily use **C++ for Data Structures, Algorithms, simulation logic, and routing**, supported by an appropriate UI technology for interactive visualization. The architecture will also be designed to allow the potential integration of **AI/ML-based traffic prediction** in a future phase.

5. System Architecture



Basic Data Flow:
User selects route → Read current traffic → Update graph edge weights → Run A* → Priority Queue explores nodes → Optimal route calculated → Display route on map → Traffic condition changes → Update graph → Recalculate route → Display new route.

6. Project Outcome / Deliverables

The project will deliver an interactive traffic routing and congestion simulation system capable of modelling a road network and finding efficient routes between selected locations.

The major deliverables will include:
• A C++ implementation of the weighted graph and supporting data structures.
• An implementation of the A* pathfinding algorithm.
• Dynamic traffic and road-condition simulation.
• Automatic route recalculation when road conditions change.
• Interactive road-network/map visualization.
• Source and destination selection through the UI.
• Visual representation of traffic levels and selected routes.
• Simulated vehicle movement.
• Route statistics such as distance, estimated travel cost/time, and nodes explored.
• Testing and analysis of algorithm performance.

7. Assumptions

The project will use a simulated road network rather than real-time city traffic data. Intersections and roads will be represented as nodes and weighted edges in a graph. Traffic levels will initially be generated or controlled through the simulation rather than obtained from live traffic APIs.

The road network will be assumed to have valid connections between locations, and the A* heuristic will be selected appropriately for the simulated map. Traffic changes will modify road weights, while a closed road will be treated as unavailable for routing.

8. References

1. C++ Documentation — Basic C++ programming, OOP, STL containers, and data structures.
2. GeeksforGeeks — References and tutorials on Graphs, A* Algorithm, Priority Queues, and Data Structures.
3. W3Schools — C++ programming and Object-Oriented Programming concepts.
4. Project Reference Material — *DSA + C++ Project Ideas – Explained*.
5. YouTube